

1. Project Proposal

Overview

The Student Performance Dashboard project aims to design and implement a data pipeline which will collect and process and store and visualize student academic performance data. The system uses Python for data preprocessing and SQL for data storage and querying and visualization tools such as Matplotlib and Streamlit to display insights.

Objectives

- Build a clean and structured dataset of student performance
- Develop a relational database to store academic data
- Enable analytical querying using SQL
- Visualize trends in student performance and attendance
- Provide a simple dashboard for decision-making

Scope

In Scope:

- CSV data ingestion and preprocessing using Python
- SQL database design and integration (SQLite/PostgreSQL)
- Data analysis using SQL queries
- Visualization using Python libraries
- Optional interactive dashboard

Out of Scope:

- Real-time data streaming
- Advanced ML predictions (future extension)
- Large-scale distributed systems

2. Project Plan

Task / Week	Week 1	Week 2	Week 3	Week 4
Project Planning				
Data Collection (CSV)				
Data Cleaning & Preprocessing				
Database Design (ERD)				
SQL Implementation				
SQL Query Development				
Data Analysis				
Visualization (Matplotlib/Seaborn)				
Dashboard Development				
Testing & Validation				
Documentation (Report)				
Presentation Preparation				

Milestones

- Milestone 1: Clean dataset ready
- Milestone 2: Database + SQL queries completed
- Milestone 3: Visualizations built
- Milestone 4: Final report & presentation

Deliverables

- Python preprocessing script
- Cleaned dataset (CSV)
- SQL schema + queries
- ER Diagram
- Visualization notebook
- Optional dashboard
- Final report (PDF)
- Presentation slides

Resource Allocation

- Tools: Python (Pandas, Matplotlib), SQL (SQLite/PostgreSQL), Streamlit
- Hardware: Standard laptop
- Human Resources: 5 team members

3. Task Assignment & Roles

Team Roles

1. Data Engineer

- Handle CSV data collection and preprocessing
- Clean and transform dataset
- Handle missing values, duplicates, formatting
- Ensure data quality and consistency

2. Database Developer

- Design relational schema (ER Diagram)
- Create SQL tables and relationships
- Normalize data structure
- Optimize database performance

3. Data Analyst

- Write SQL queries for insights
- Analyze performance trends (scores, attendance)
- Generate aggregated metrics (averages, top students)
- Prepare processed data for visualization

4. Visualization Engineer

- Create charts using Matplotlib / Seaborn
- Design clear and readable graphs
- Build dashboard using Streamlit
- Improve UI/UX of the dashboard

5. Project Manager / QA Engineer

- Track progress and ensure deadlines are met
- Coordinate between all team members
- Review outputs (data, SQL, visuals) for correctness
- Perform testing & validation
- Prepare final report and presentation

4. Risk Assessment & Mitigation Plan

Risk	Impact	Likelihood	Mitigation
Poor data quality (missing/duplicates)	High	Medium	Apply preprocessing & validation rules
SQL errors / poor schema design	Medium	Medium	Review ERD before implementation
Delays in milestones	High	Medium	Weekly progress tracking
Visualization bugs	Medium	Low	Test with sample datasets
Team coordination issues	Medium	Medium	Use task tracking tools (Trello/Notion)
Lack of dashboard experience	Low	Medium	Use simple Streamlit templates

5. KPIs (Key Performance Indicators)

Technical KPIs

- Data Accuracy Rate: $\geq 95\%$ clean data after preprocessing
- Query Execution Time: < 2 seconds
- System Uptime (if dashboard): $\geq 95\%$

Project KPIs

- Milestone Completion Rate: 100% on time
- Bug Rate: $< 5\%$ critical issues

User/Impact KPIs

- Dashboard Usability: Easy to interpret insights
- Insight Generation: At least 3–5 meaningful patterns discovered